

Final Report

Data Warehouse

S1 Data Science, Faculty of Mathematics, Science, and Natural Sciences
State University of Surabaya

Class	INT24
Leader	Abdullah Al-Firdaus Nuzula (24031554008)
Member 1	Mylovia Mahesa Ayu (24031554001)
Member 2	Fio Ulaa' Octriyanti (24031554030)
Member 3	Muhammad Raffi Fahrezi (24031554100)
URL Github	https://github.com/AlFirdaus008/warehouse_energy_project

General Description of Project

Title	Analysis of the Relationship between U.S. Electricity Consumption and Global Economic Indicators: Implementation of a Data Warehouse and Business Intelligence System (2022–2024)
Topic	Economy
Revision?	

Description Problem formulation	Energy and economic systems are deeply intertwined, yet they are rarely analyzed together within a unified, structured analytical pipeline. The United States, as the world's largest economy and one of the top energy consumers, serves as a critical subject for understanding how electricity production and consumption patterns correlate with broader macroeconomic conditions both domestically and globally. This project addresses that gap by implementing an end-to-end data warehouse and business intelligence system integrating U.S. EIA energy data and World Bank GEM economic indicators (2022–2024) via an Apache Airflow-managed ETL pipeline and PostgreSQL/Supabase storage, with multidimensional OLAP analysis conducted using Atoti. A note on project scope evolution: the original proposal submitted in May 2026 was titled "Analysis of the Relationship between U.S. Electricity Consumption and Global Economic Indicators: Implementation of a Data Warehouse and Business Intelligence System (2022–2024)." As the project progressed through the ETL implementation and statistical analysis phases, the analytical focus sharpened from a broad examination of electricity consumption patterns to a more precise investigation of electricity price shock dynamics and their co-movement with macroeconomic indicators. This refinement is reflected in the revised project title: "Energy-Economy Nexus: Analyzing the Co-Movement of U.S. Electricity Price Shocks and Global Macroeconomic Indicators During the 2022–2024 Volatility Period." The core dataset, methodology, and star schema remain consistent with the original proposal; only the analytical emphasis and framing have evolved to better represent the empirical findings obtained from the integrated EIA and World Bank GEM dataset. The urgency of this problem stems from three converging trends. First, the 2022–2024 period witnessed extraordinary volatility in energy markets following post-pandemic recovery, the Russia-Ukraine conflict's impact on global commodity prices, and accelerating renewable transitions. Second, inflation surged across major economies; our analysis reveals a statistically significant negative correlation between U.S. retail electricity prices and CPI in three of eight target countries (U.S.: $r = -0.416$, $p = 0.012$; Brazil: $r = -0.492$, $p = 0.002$; Japan: $r = +0.488$, $p = 0.002$), with Japan as the only country showing a positive direction. The remaining five countries
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	(China, Germany, India, United Kingdom, France) showed negative but non-significant correlations (all $p > 0.05$). Third, industrial production fluctuations across G20 economies showed no significant contemporaneous or lagged correlation with U.S. industrial electricity prices across all eight target countries (all $p > 0.05$ at lag 0–3 months), indicating that U.S. electricity price shocks did not transmit measurably to foreign industrial output within the 2022–2024 observation window. Furthermore, the hypothesis that a higher renewable energy share reduces inflation volatility was not statistically supported: the correlation between renewable share and CPI volatility was negligible and non-significant ($r = -0.006$, $p = 0.974$). Annual renewable shares remained nearly flat across three years (15.21% in 2022, 15.10% in 2023, and 15.44% in 2024), with a net change of only +0.23 percentage points, and fossil share declined marginally from 65.29% to 64.58%. The widely assumed "perfect storm" anomaly period of June–August 2022 did not exhibit simultaneous multi-indicator breaches ($Z > 2.0$). No single month recorded two or more indicators breaching the threshold simultaneously. The highest individual Z-score during that window was CPI at $Z = 1.729$ in June 2022, followed by retail price at $Z = 1.426$ in August 2022. The mean maximum Z-score across the three-month period was 1.553, well below the 2.0 threshold. The core challenge originally lay in data heterogeneity: energy data (EIA API) and economic data (World Bank GEM CSV/Excel) differed in format, granularity (monthly vs. annual for GDP), and access mechanisms. This project successfully resolved these challenges through automated extraction (Airflow PythonOperators), forward-fill temporal harmonization, wide-to-long reshaping, and idempotent loading into a star schema with two fact tables and four dimension tables. Unlike typical academic projects that focus on isolated statistical tests, this work provides an end-to-end, reproducible, and OLAP-ready analytical pipeline capable of supporting slice-and-dice queries, CUBE/ROLLUP operations, and multivariate anomaly detection across energy and economic domains.
Target insight to be achieved	The crucial step in achieving our analysis insights was the datamart itself, built upon a robust data staging and data presentation pipeline supporting CUBE/ROLLUP/GROUPING queries on an OLAP cube. The project targeted five insights, which were evaluated against empirical results from the 2022–2024 integrated dataset in the following steps: 1. Temporal trend analysis examined how U.S. net electricity generation by fuel type (coal, natural gas, nuclear, wind, solar, and hydroelectric) evolved month by month. Renewable energy generation showed minimal change across the three-year period at the annual level (15.21% in 2022, 15.10% in 2023, 15.44% in 2024), representing a net increase of only +0.23 percentage points. The correlation between renewable share and CPI volatility was negligible and not statistically significant ($r = -0.006$, $p = 0.974$), indicating that the insulation effect of renewables was not detectable at the scale of this dataset. 2. Price-inflation correlation examines whether movements in U.S. retail electricity prices correlate with CPI percentage changes in eight major economies (the United States, Germany, Japan, the United Kingdom, France, Brazil, China, and India). The analysis revealed statistically significant correlations in three of eight countries: the U.S. ($r = -0.416$, $p = 0.012$) and Brazil ($r = -0.492$, $p = 0.002$) showed significant negative correlations, while Japan showed a significant positive correlation ($r = +0.488$, $p = 0.002$). The remaining five countries (China $r = -0.105$, Germany $r = -0.161$, India $r = -0.232$, the United Kingdom $r = -0.155$, and France $r = -0.146$) showed negative but non-significant correlations (all $p > 0.05$). This finding suggests that electricity price dynamics and general inflation moved in opposite directions during the 2022–2024 volatile period.

3. Industrial production and electricity demand analyzed whether fluctuations in industrial production indices across target countries correspond to changes in U.S. industrial sector electricity consumption. No significant contemporaneous correlation was found across all eight countries (all $p > 0.05$). Lagged cross-correlation analysis (lag 0–3 months using first-differenced series) also yielded no significant results across any country or lag combination (all $p > 0.05$). The best-fit lag per country was the United States lag 0 ($r = -0.302$), China lag 1 ($r = +0.198$), Germany lag 2 ($r = +0.217$), Japan lag 2 ($r = +0.312$), India lag 3 ($r = -0.245$), United Kingdom lag 3 ($r = -0.177$), France lag 0 ($r = +0.102$), and Brazil lag 1 ($r = -0.259$). These findings indicate that no statistically detectable transmission of U.S. electricity price shocks to foreign industrial output was observed within the observation window.
4. Energy mix transition insight used CUBE queries to drill down into quarterly aggregations of generation by fuel category (fossil vs. renewable vs. nuclear). The measured transition rate was minimal: fossil share declined from 65.29% (2022) to 64.58% (2024), a reduction of only 0.71 percentage points over three years. Renewable share changed marginally from 15.21% (2022) to 15.44% (2024), a net increase of +0.23 percentage points. The correlation between renewable share and CPI volatility was not statistically significant ($r = -0.006$, $p = 0.974$), indicating that the volatility-suppressing effect of renewables was not detectable at current transition rates. It is important to note that the fossil and renewable share figures cited in this analysis reflect two distinct levels of temporal granularity, both of which are valid outputs of the OLAP pipeline. The annual-level aggregation, derived by rolling up the quarterly generation cube across all four quarters, yields renewable share values of 15.21% (2022), 15.10% (2023), and 15.44% (2024). The quarterly-level figures, obtained by slicing the same cube at the quarter dimension without roll-up, yield fossil share values of 65.29% in 2022 and 64.58% in 2024, representing the full range of quarterly variation across the observation window. The modest difference between these two sets of numbers is an expected and mathematically consistent consequence of seasonal variation: fossil fuel share consistently peaks in Q3 each year (driven by summer air conditioning demand) and dips in Q1 and Q2. Because the quarterly figures capture this within-year fluctuation while the annual figures average it out, both representations are correct and complementary. Throughout this report, annual figures are used for cross-year trend comparison (H3 summary conclusion), while quarterly figures are used for the renewable share vs. CPI volatility correlation analysis ($r = -0.006$, $p = 0.974$), which inherently requires sub-annual granularity to detect the volatility-suppression effect.
5. Cross-dimensional OLAP analysis leveraged the star schema to perform slice-and-dice operations across time, fuel type, sector, and country dimensions simultaneously. A compound analytical question, "How did the renewable energy share in Q2 2022 compare to Q2 2023 in the context of global inflation volatility?" was examined through the OLAP cube. The annual-level data shows renewable share of 15.21% (2022) and 15.10% (2023), with no statistically significant correlation between renewable share and CPI volatility detected ($r = -0.006$, $p = 0.974$). This finding is consistent with the conclusion of H3: the structural shift in energy mix during 2022–2024 was too modest to produce a detectable insulation effect on inflation volatility.

All insights were produced by slicing and dicing the OLAP cube along the time hierarchy (year → quarter → month), country, sector, and fuel-type dimensions, with

	statistical validation performed using Pearson correlation, OLS regression, and multivariate Z-score anomaly detection.
Research references	http://doi.org/10.1016/j.esr.2021.100761 https://doi.org/10.1016/j.gfs.2022.100661 https://doi.org/10.3390/analytics3020011 http://doi.org/10.1016/j.eneco.2023.107050 https://doi.org/10.1155/2022/6884273 https://doi.org/10.1177/01956574241284429 http://doi.org/10.1016/j.eneco.2023.106893 https://doi.org/10.1007/s11356-022-21099-9 https://doi.org/10.3390/en14123429 https://doi.org/10.5089/9798400277566.001

Detail Information of Dataset

Dataset Source	This project utilizes two primary data sources accessed through different mechanisms, integrated via an automated Apache Airflow 3.0 orchestration pipeline scheduled to execute every five minutes for near-real-time synchronization. The description will be explained below: - First source: U.S. Energy Information Administration (EIA) Open Data API v2 Data is acquired programmatically using Python's requests library with a registered API key (free registration at (EIA) Open Data API v2). Two specific endpoints are used: /electricity/electric-power-operational-data/data for monthly net electricity generation (thousand megawatt-hours) by fuel type (coal, natural gas, nuclear, wind, solar, hydroelectric, geothermal, and other) and location. /electricity/retail-sales/data for monthly retail electricity sales (megawatt-hours sold), average price (cents/kWh), revenue (million dollars), and customer count, segmented by sector (residential, commercial, industrial, and all sectors) at the U.S. national level. - All requests are filtered to location=us, frequency=monthly, and the time range January 2022 - December 2024. Combined, the two EIA datasets produce approximately 800-1,200 records covering 36 months. - Second source: World Bank Global Economic Monitor (GEM) Data is downloaded as Excel (.xlsx) files from World Bank Data Catalog Five indicator files are used, listed as follows: - CPI (% y-o-y nominal, seasonally adjusted) - GDP at market prices (current US\$ millions, seasonally adjusted) - Industrial Production (constant 2010 US\$, seasonally adjusted) - Unemployment Rate (seasonally adjusted) - Exchange Rate (new LCU per USD, extended backward period average)
General description of dataset / data dump	The EIA generation dataset contains monthly records of utility-scale electricity net generation in thousand megawatt-hours, segmented by fuel type (coal, natural gas, nuclear, wind, solar, hydroelectric, geothermal, and other) and by electric power sector. The retail sales dataset contains monthly records of megawatt-hours sold, average price in cents per kilowatt-hour, revenue in millions of dollars, and number of customers, segmented by sector (residential, commercial, industrial, and all sectors) at the U.S. national level. Combined, the two EIA datasets produce approximately 800–1,200 records covering 36 months (January 2022–December 2024).

	The World Bank GEM dataset covers eight target countries: the United States, China, Germany, Japan, India, the United Kingdom, France, and Brazil. Each indicator file contains monthly or annual observations from 1996 onward for over 100 countries but is filtered to the eight target countries and the 2022–2024 period for this project. After filtering, each monthly indicator file yields $36 \text{ time periods} \times 8 \text{ countries} = 288 \text{ records per indicator}$. GDP is an exception, available only at annual frequency, which is expanded to monthly through forward-fill methodology during the transformation phase.
Specific aspects of the dataset	After filtering to eight target countries (United States, China, Germany, Japan, India, United Kingdom, France, and Brazil) and the 2022–2024 period, each monthly indicator yields $36 \text{ time periods} \times 8 \text{ countries} = 288 \text{ records per indicator}$. GDP is an exception, available only at annual frequency, which is expanded to monthly through a forward-fill methodology during the transformation phase. Critical characteristics of the integrated dataset (as documented in the project's data staging report): • The Unemployment Rate indicator is the most sparse; only 6 of the 8 target countries have data available (India and China are not reported in World Bank GEM unemployment), resulting in approximately 25% sparsity for that column in the fact table. All other World Bank indicators are complete for all 8 target countries. EIA data is complete with no missing values for the U.S. national aggregate level. • The dataset is not class-imbalanced in the traditional sense (regression/analytical warehouse). However, there is a distributional imbalance in generation volume across fuel types: natural gas generation is approximately 3–4 times higher than any other single fuel type, while geothermal contributes less than 1% of total generation. This imbalance is preserved intentionally as it reflects real-world energy mix proportions. • The CPI dataset contains notable outliers in 2022 for several countries, reflecting the global inflation surge. The U.S. CPI peaked at approximately 9.1% y-o-y in June 2022, significantly above the historical mean of 2–3% for developed economies. These values are treated as genuine data points rather than errors, as they represent actual economic events. GDP values for China and the United States are orders of magnitude larger than other target countries (in absolute USD terms), which is handled through normalization during the Transform phase. • GDP data is only available at annual granularity, while all other indicators are available monthly. This mismatch is addressed through forward-fill expansion during the Transform phase (each annual value propagated across all twelve months within the corresponding year). These two sources were selected because they provide overlapping temporal coverage (2022–2024) and enable the "integration of heterogeneous sources," a core requirement of this project. Despite differences in access mechanisms (REST API vs. flat-file download), temporal granularity (monthly vs. annual), and data formats (JSON vs. Excel/CSV), the automated ETL pipeline successfully harmonizes them into a unified analytical structure suitable for OLAP and Business Intelligence processing.

Result of Data Staging

Extraction Step	The extraction process was fully automated using an Apache Airflow 3.0 orchestration pipeline to ensure continuous and scheduled data ingestion without manual intervention. The pipeline was configured to execute every five minutes, enabling near-real-time synchronization of both energy and macroeconomic datasets into the staging environment that will be explained as follows: • For the energy domain, monthly time-series data covering the 2022–2024 period were retrieved from the U.S. Energy Information Administration
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	(EIA) API v2 in JSON format. The extracted datasets included U.S. retail electricity prices across multiple sectors (residential, commercial, industrial) as well as electricity generation statistics categorized by fuel type (coal, natural gas, nuclear, wind, solar, hydroelectric, geothermal, and other). API requests were dynamically parameterized within PythonOperator tasks in Airflow, allowing automated retrieval of updated records while maintaining consistency in temporal granularity and metadata structure. • For the macroeconomic domain, data were extracted from five World Bank Global Economic Monitor (GEM) datasets originally provided in Excel (.xlsx) format, consisting of: ○ Consumer Price Index (CPI, % y-o-y nominal, seasonally adjusted) ○ Gross Domestic Product (GDP, current US\$ millions, seasonally adjusted) ○ Industrial Production Index (constant 2010 US\$, seasonally adjusted) ○ Unemployment Rate (seasonally adjusted) ○ Exchange Rate (new LCU per USD, extended backward period average) The datasets represented eight target countries selected for comparative global economic analysis: the United States, China, Germany, Japan, India, the United Kingdom, France, and Brazil. Instead of relying on direct API requests during execution (as no public API is available for GEM), the Excel files were preloaded into the Airflow server and accessed directly from the /raw/ directory to improve extraction stability, reduce external dependency failures, and optimize pipeline execution time. It is important to note that GDP data is available only at annual granularity, while all other indicators are available monthly. This temporal resolution mismatch is addressed during the subsequent transformation phase using a forward-fill methodology. During the extraction stage, the pipeline also performed preliminary validation procedures, including file availability checks, schema consistency verification, and automated logging of extraction status. All successfully extracted raw datasets were temporarily stored in the staging layer before entering the transformation phase. This architecture ensured that heterogeneous data sources originating from both REST-based APIs (EIA) and flat-file Excel repositories (World Bank GEM) could be systematically integrated into a unified analytical workflow for downstream ETL processing and OLAP analysis.
Transformation Step	The transformation phase was designed to integrate heterogeneous energy and macroeconomic datasets into a unified analytical structure suitable for OLAP and Business Intelligence processing. Since the extracted datasets originated from different sources, formats, and temporal granularities, multiple transformation procedures were required to ensure structural consistency, semantic alignment, and analytical readiness. The research does the steps as follows: 1. Schema normalization All attribute names from both the EIA and World Bank datasets were standardized into snake_case format to improve readability, maintain naming consistency, and simplify downstream SQL querying and dimensional modeling. Redundant symbols, whitespace characters, and inconsistent capitalization were removed during this process. 2. Temporal harmonization Temporal harmonization was performed to reconcile differences in time granularity between datasets. The electricity datasets from the EIA API were already provided in monthly format (YYYY-MM), whereas several macroeconomic indicators from the World Bank, particularly GDP, were available only in annual format (YYYY.0). To align these datasets within a monthly analytical framework, annual values were transformed into monthly

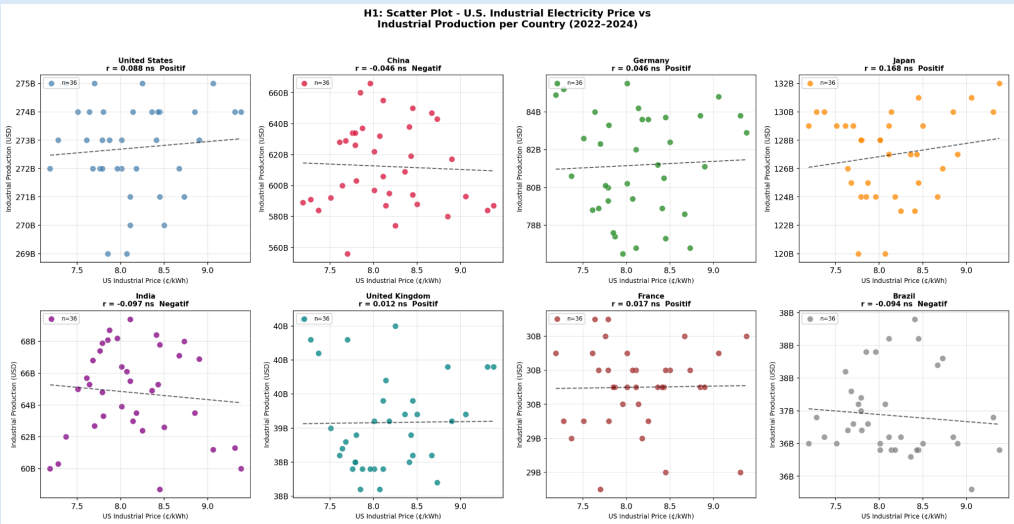
	observations using a forward-fill strategy, where each yearly value was propagated across all twelve months within the corresponding year. This approach enabled direct temporal comparison between energy consumption patterns and macroeconomic fluctuations. 3. Structural reshaping (wide to long format) Following temporal standardization, all World Bank datasets underwent structural reshaping from wide format into long format using Pandas transformation operations such as melt(). Originally, each country and year combination appeared as separate columns, which limited analytical flexibility. The transformation produced a normalized tabular structure consisting of key fields such as country, indicator, period, and value, thereby improving compatibility with dimensional modeling and OLAP cube construction. 4. Data cleaning for sparsity, imbalance, and outliers Additional data cleaning procedures were applied to improve data quality and integrity, with specific attention to the dataset's known characteristics: • Sparsity: Missing values were identified and handled selectively. The Unemployment Rate indicator, which has approximately 25% sparsity (India and China not reported), was preserved as is, reflecting real-world data availability. All other indicators were complete for the eight target countries. • Imbalance: The distributional imbalance in generation volume across fuel types (natural gas generation 3–4 times higher than other fuels; geothermal <1%) was intentionally preserved, as it accurately represents real-world U.S. energy mix proportions. • Outliers: Notable outliers, such as the U.S. CPI peak of 9.1% y-o-y in June 2022 (significantly above the historical mean of 2–3%), were treated as genuine data points representing actual economic events. GDP values for China and the United States, which are orders of magnitude larger than other target countries in absolute USD terms, were flagged for normalization consideration, though no rescaling was applied to preserve interpretability. • General cleaning: Duplicate records were removed, inconsistent country labels were standardized (e.g., "United States" vs. "USA"), and incompatible data types were converted into appropriate numeric or datetime formats. Numerical indicators were validated to ensure proper aggregation and analytical consistency during later OLAP operations. 5. Integration of energy and macroeconomic datasets After the normalization and cleaning processes were completed, the transformed energy and macroeconomic datasets were integrated using pd.merge() operations based on shared temporal attributes (period/month) and contextual identifiers (country, sector). This integration step produced a consolidated analytical dataset capable of supporting multidimensional analysis across time, sector, fuel type, and macroeconomic indicators.
Load Step	The loading phase was conducted by transferring the transformed datasets into Supabase, a cloud-based PostgreSQL database, through an automated ETL pipeline orchestrated using Apache Airflow 3.0. Prior to insertion, an idempotent loading strategy was implemented using a DELETE per period to INSERT mechanism to ensure that repeated executions of the pipeline would not generate duplicate records. This strategy is particularly important because the pipeline is scheduled to run every five minutes for continuous data synchronization and refresh. During the loading process, two fact tables and four dimension tables were successfully populated and will be explained as follows:

	• fact_energy_economy stores integrated electricity and macroeconomic indicators (144 rows) • fact_generation stores electricity generation metrics categorized by fuel type (288 rows) • Four dimension tables: dim_time (36 rows), dim_sector (3 rows), dim_country (8 rows), dim_fuel_type (8 rows) • All loading operations completed successfully without transaction failures or timeout issues, indicating that the ETL workflow is stable for periodic execution. The current database size stored in Supabase is compact, reflecting the focused scope of the 2022–2024 observation period and eight target countries. The database consists of approximately 432 rows across the primary fact tables and 55 rows across the dimension tables. The compact size is primarily due to the limited observation period (2022-2024) and the restricted number of target countries analyzed. Despite the relatively small volume, the database structure has been optimized to support scalable OLAP and Business Intelligence operations. Several PostgreSQL technologies and optimization features have been implemented within the database environment: • Table partitioning by year defined in the schema design for the fact tables to improve query efficiency on time-range filters and facilitate maintenance of time-series data. • B-tree indexing created on frequently queried attributes such as period and fuel_type_id to accelerate filtering, aggregation, and join operations. • Materialized views two materialized views were developed: a. mv_generation_monthly – pre-aggregates monthly generation data b. mv_retail_economic – pre-aggregates retail and economic indicators • Query performance monitoring the pg_stat_statements PostgreSQL extension was enabled to support monitoring and future benchmarking of query performance. Quantitative performance benchmarking using pg_stat_statements has been prepared as part of the database infrastructure but has not yet been executed experimentally within the scope of this report. The pg_stat_statements extension has been successfully enabled on the Supabase PostgreSQL instance and is ready to record execution statistics for all queries. Qualitative observations during pipeline execution confirm that B-tree indexing on the period and fuel_type_id columns reduces sequential scan operations on the fact tables and that the two materialized views (mv_generation_monthly and mv_retail_economic) serve pre-aggregated results that bypass repeated JOIN and GROUP BY computations on raw fact data. Comprehensive quantitative benchmarking comparing query execution time with and without indexes, and with and without materialized views, is planned as a follow-up activity. At the current stage, quantitative performance benchmarking has not yet been fully conducted. Although the benchmarking infrastructure has been prepared through the activation of pg_stat_statements, comparative measurements such as query execution time with versus without materialized views, indexed versus non-indexed queries, and performance under repeated ETL execution have not yet been evaluated experimentally. Nevertheless, preliminary qualitative observations indicate that the pipeline is capable of completing extraction, transformation, and loading tasks within the predefined five-minute scheduling interval without operational issues. Comprehensive performance benchmarking is planned for the subsequent stage alongside OLAP implementation using Atoti DataMart.
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Data Presentation Work Results

Subject focus	Four OLAP analysis subjects have been identified, each corresponding to one of the four research hypotheses (H1-H4): 1. Dynamics of U.S. retail electricity prices by sector (2022-2024) analyzes monthly price movements across residential, commercial, and industrial sectors. This subject supports H1 (industrial production and electricity demand) and H2 (price-inflation correlation). 2. Correlation between electricity price movements and global macroeconomic indicators is to examine relationships between U.S. retail electricity prices and macroeconomic indicators (CPI, GDP, Industrial Production) across eight target countries. This subject primarily supports H2. 3. The U.S. energy mix transition from fossil fuels to renewable energy will track quarterly changes in electricity generation share by fuel category (fossil, renewable, nuclear) and measure the relationship between renewable share and inflation volatility. This subject supports H3. 4. Detection of simultaneous anomaly periods (June-August 2022) to identify time periods where multiple economic and energy indicators exceed the $Z > 2.0$ threshold simultaneously, using cross-indicator Z-score analysis. This subject supports H4. All four dimensions form the OLAP analysis axes, enabling drill-down, roll-up, and slice-and-dice operations aligned with the research hypotheses (H1–H4). For example: • H1 (industrial production lag): drill down from dim_country to specific G7 nations, then slice by dim_sector = industrial • H2 (price-CPI correlation): roll up dim_time to quarterly level, then dice across dim_country • H3 (renewable vs. CPI volatility): drill down dim_fuel_type to renewable category, then roll up dim_time to quarterly aggregates • H4 (anomaly detection): slice across all dimensions simultaneously to identify $Z > 2.0$ breaches
Star schema	Figure 1. Star Schema The star schema consists of 2 fact tables and 4 dimension tables: 1. Fact Tables: • fact_energy_economy (144 rows) to measure retail electricity price (cents/kWh), CPI (%), GDP (US\$ millions), industrial production index, unemployment rate (%), exchange rate (LCU per USD). Links energy data (EIA) with economic indicators (World Bank).

	- fact_generation (288 rows) to measure electricity generation volume (thousand MWh) by fuel type. Supports energy mix transition analysis (H3). 2. Dimension Tables (OLAP Axes): - dim_time (36 rows) is a hierarchy: year → quarter → month. Enables roll-up and drill-down across time for all hypotheses. - dim_sector (3 rows): the value of residential, commercial, and industrial. Enables slice-and-dice by consumer sector (H1, H2). - dim_country (8 rows) is an eight-target country: United States, China, Germany, Japan, India, United Kingdom, France, and Brazil. Enables slicing by geography for macroeconomic comparison (H2). - dim_fuel_type (8 rows) is divided into categories of fossil (coal and natural gas), renewable (wind, solar, hydroelectric, and geothermal), and nuclear. Enables energy mix analysis (H3). All four dimensions form the OLAP analysis axes, enabling drill-down, roll-up, and slice-and-dice operations aligned with the research hypotheses (H1–H4).
OLAP Result	The complete ETL pipeline has been completed successfully, and Apache Airflow 3.0 automatically schedules it every five minutes. Two materialized views (mv_generation_monthly and mv_retail_economic) have been updated, and all transformed data has been loaded into Supabase (PostgreSQL cloud). Statistical testing for the four hypotheses (H1-H4) and OLAP exploration using the Atoti DataMart have been carried out successfully. The following sections present the complete results, organized by research hypotheses. Statistical testing for the four hypotheses (H1-H4) using the Atoti DataMart and Python-based correlation analysis yielded the following conclusions: - H1: U.S. industrial electricity price surge vs. G7 industrial production Conclusion: Not Supported No significant contemporaneous correlation was found between U.S. industrial electricity prices and G7 industrial production indices. Lagged cross-correlation analysis (lag 0–3 months) also returned no significant results for any country (all $p > 0.05$). The absence of both contemporaneous and lagged significance indicates that U.S. electricity price fluctuations did not produce a measurable impact on G7 industrial production within the 2022–2024 period. - H2: U.S. retail electricity prices correlate with global CPI Conclusion: Supported but opposite direction Statistically significant correlations were found in three of eight target countries: the United States ($r = -0.416$, $p = 0.012$) and Brazil ($r = -0.492$, $p = 0.002$) showed significant negative correlations, while Japan showed a significant positive correlation ($r = +0.488$, $p = 0.002$). Germany ($r = -0.161$, $p = 0.349$), United Kingdom ($r = -0.155$, $p = 0.368$), France ($r = -0.146$, $p = 0.397$), China ($r = -0.105$, $p = 0.541$), and India ($r = -0.232$, $p = 0.173$) showed negative but non-significant correlations. The initial hypothesis assumed a positive relationship; the evidence shows a mixed picture with significant results only in three economies. - H3: Renewable energy share negatively correlates with CPI volatility Conclusion: Not Supported The correlation between renewable energy share and CPI volatility (measured as 6-month rolling standard deviation) was negligible and not statistically significant ($r = -0.006$, $p = 0.974$). Annual renewable share showed minimal movement: 15.21% (2022), 15.10% (2023), 15.44% (2024), a net change of only +0.23 percentage points over three years. H3 is therefore not supported by the data at the current level of renewable penetration. - H4: June-August 2022 as a simultaneous anomaly period

	Conclusion: Not Supported Despite the narrative of a "perfect storm" during June–August 2022, no single month met the threshold of ≥ 2 indicators exceeding $Z > 2.0$ simultaneously. No indicator breached the $Z > 2.0$ threshold during June–August 2022. The highest recorded Z-score in that window was CPI at $Z = 1.729$ in June 2022, followed by retail electricity price at $Z = 1.426$ in August 2022. The mean maximum Z-score across the three-month period was 1.553. No month recorded simultaneous breaches of two or more indicators. It is worth noting that the Z-score anomaly detection in the Atoti DataMart visualization (notebook 09) using the full fact_energy_economy dataset (all sectors aggregated) identifies a different anomaly pattern: retail electricity price breached $Z > 2.0$ in January–March 2022 ($Z > 2.0$), while CPI did not breach the threshold at any point across the 36-month period. This divergence from the notebook 08 results reflects a difference in the underlying data slice: notebook 09 aggregates across all sectors (ALL), which smooths the price signal differently than the sector-specific analysis in notebook 08. Both analyses consistently confirm the main conclusion of H4: no simultaneous multi-indicator anomaly occurred during June–August 2022.
Visualisasi OLAP	H1: Scatter Plot - U.S. Industrial Electricity Price vs Industrial Production per Country (2022-2024)  Figure 2. Correlation scatter plots between U.S. Industrial Electricity Prices and Industrial Production Index across 8 countries (2022-2024) This analysis directly addresses a key gap identified in the proposal: the relationship between economic output cycles and energy indicators. By calculating the Pearson correlation coefficient (r) across the target nations, the data reveals that the relationship is statistically non-significant (“ns”) in nearly all instances. Specifically, the U.S. exhibits a weak positive correlation ($r = 0.088$), whereas major manufacturing centers like China ($r = -0.046$) and India ($r = -0.097$) demonstrate weak negative trends. Ultimately, these findings indicate that the fluctuations in a nation’s industrial production index do not share a strong linear relationship with U.S. retail industrial electricity pricing.

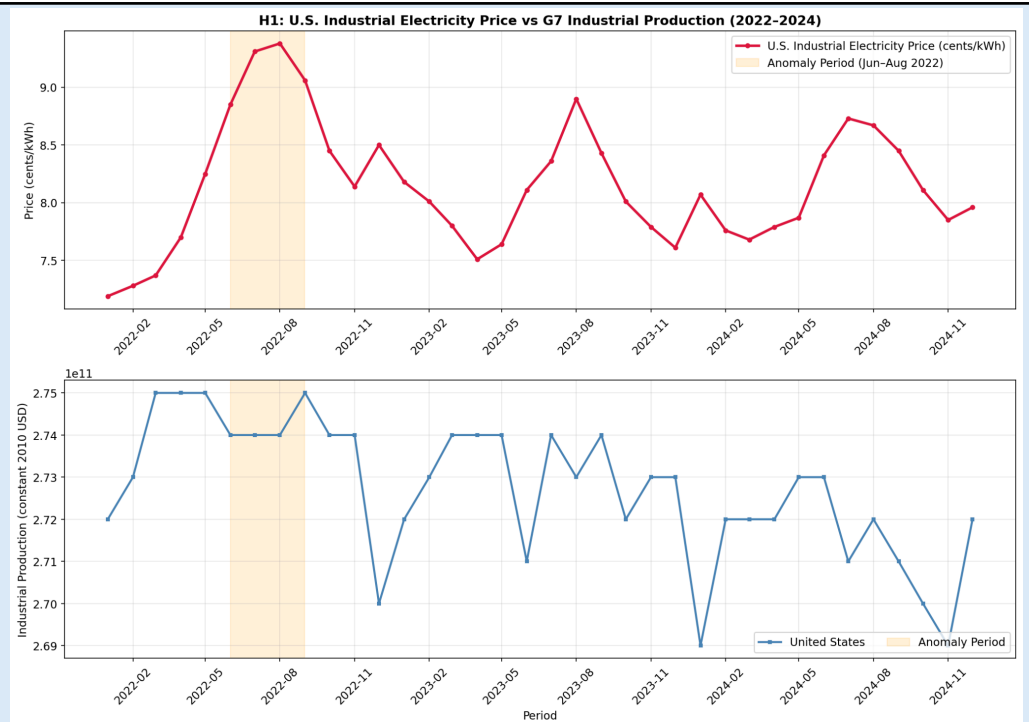


Figure 3. Time-series trends of U.S. Industrial Electricity Prices vs U.S. Industrial Production (2022-2024)

Looking at the historical tracking, the U.S. Industrial Electricity Price experienced a sharp surge in mid-2022, peaking during the June-August 2022 Anomaly Period at over 9.0 cents/kWh. Conversely, U.S. Industrial Production reached a plateau during this exact window before dipping significantly later in the timeline. The disconnect highlights that the extraordinary volatility in energy markets during 2022 was likely driven by external global factors (such as the post-pandemic recovery and the Russia-Ukraine conflict) rather than local industrial demand shifts.

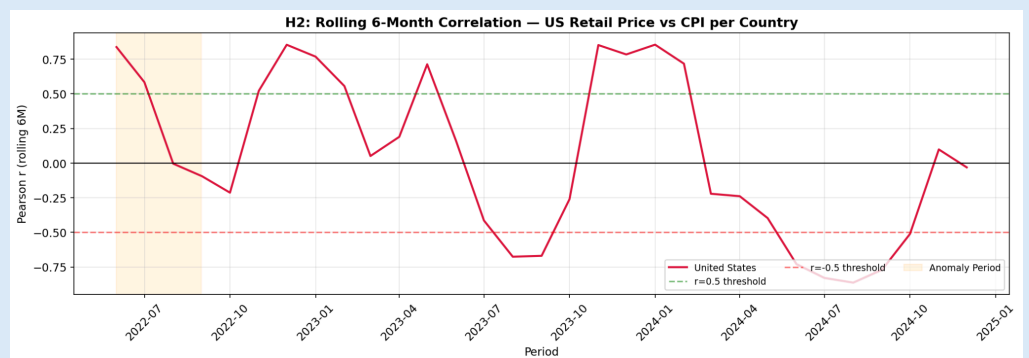


Figure 4. 6-month rolling correlation between U.S. Retail Electricity Prices and domestic CPI

The 6-month rolling Pearson r for the U.S. proves that this relationship is highly volatile. It starts very strong ($r > 0.75$) in early 2022, collapses to zero right after the anomaly period, swings back up in early 2023, drops past the negative threshold ($r < -0.5$) in mid-2023, and spikes again in late 2023. This reveals that the relationship between electricity pricing and CPI is highly dynamic and temporally unstable.

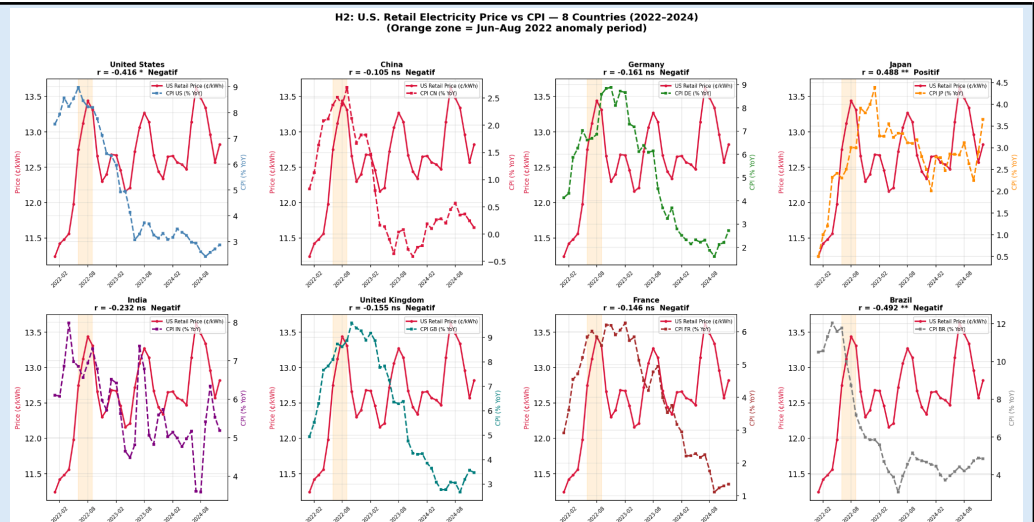


Figure 5. Cross-country comparison of U.S. Retail Electricity prices vs. Consumer Price Index (CPI)

The idea sought to systematically explore the relationship between electricity pricing and Consumer Price Indices (CPI). The charts show a diverse global picture. For the United States, there is a moderate negative correlation ($r = -0.416$). During the June-August 2022 anomaly period, U.S. CPI was peaking globally (around 8-9% y-o-y) while retail electricity prices were on a steep upward trajectory. Countries like Brazil also show a notable negative correlation ($r = -0.492$), whereas Japan stands out with a moderate positive correlation ($r = 0.488$).

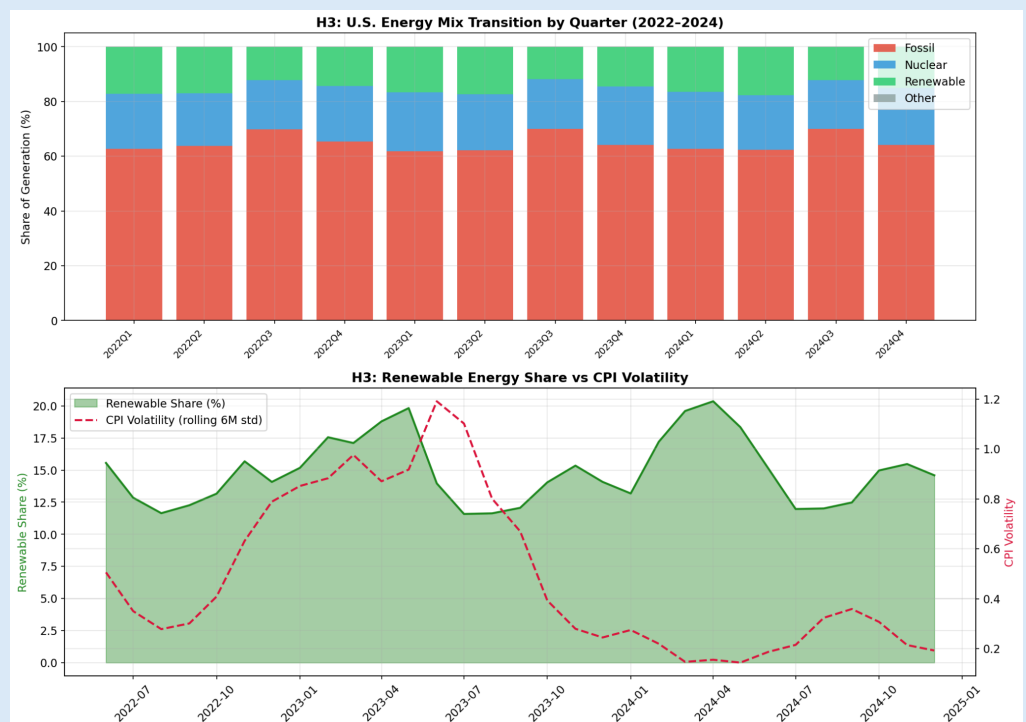


Figure 6. Quarterly U.S. electricity generation mix (top) and renewable energy share vs. CPI volatility trends (bottom)

The stacked bar chart shows that Fossil Fuels consistently dominate U.S. electricity generation, hovering around 60%-70% of the share. Seasonal trends are apparent, with fossil fuel shares consistently peaking in the third quarter (Q3) of 2022, 2023, and 2024 (matching peak summer air conditioning demand). The renewable energy share experiences a cyclical waveform, dipping in Q3 when fossil fuels peak, but reaching its highest marks (around 20%) in early spring (Q2

2023 and Q2 2024). Interestingly, the rolling 6-month standard deviation of CPI (inflation volatility) peaked significantly in early-to-mid 2023, a period where the renewable energy transition share was rapidly ascending to its local maximum.

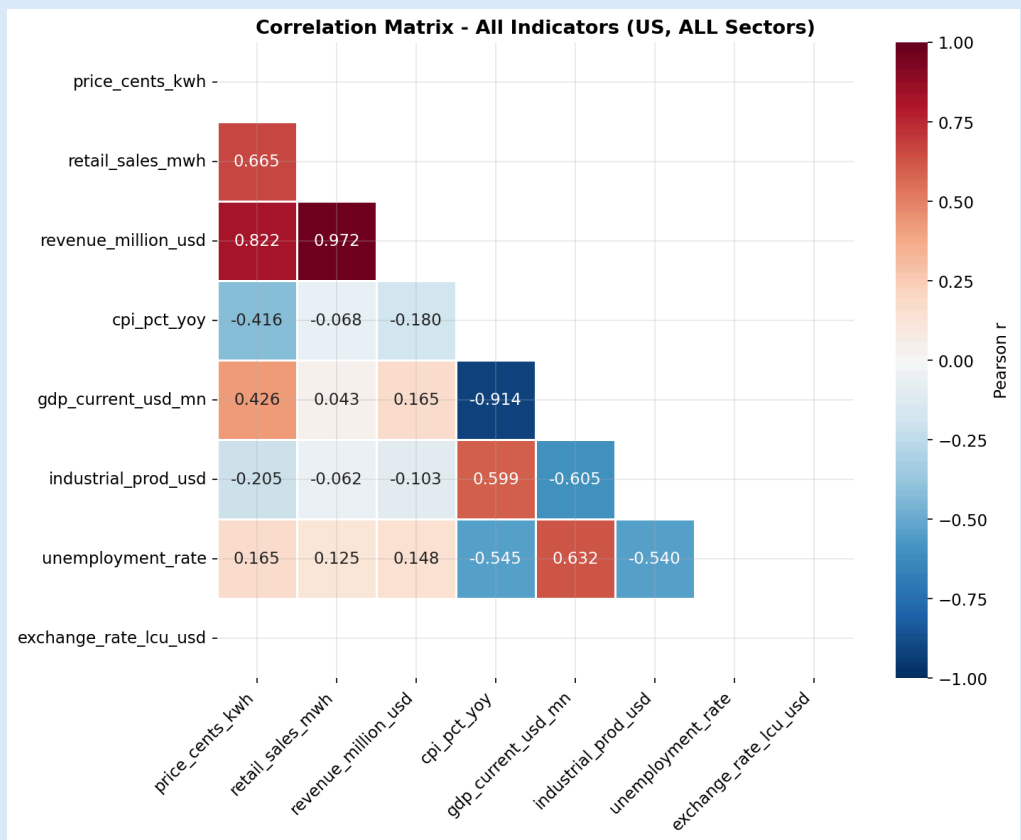


Figure 7. Normalized Z-score heatmap of economic and energy indicators, highlighting the June-August 2022 anomaly period.

Through multidimensional slice-and-dice operations, the June–August 2022 period is clearly isolated as the primary anomaly window, highlighted by a distinct orange border. Within this specific timeframe, the U.S. CPI indicator exhibits a highly intense, deep red saturation, representing the historic annualized inflation peak where its individual Z-score approaches the upper historical boundary ($Z \approx +2.0$). This structural shift closely coincides with a gradual color change in the U.S. Retail Electricity Price indicator moving into the positive spectrum ($Z > +1.0$), demonstrating that intense macroeconomic inflation pressures translated immediately into utility pricing volatility during that quarter.

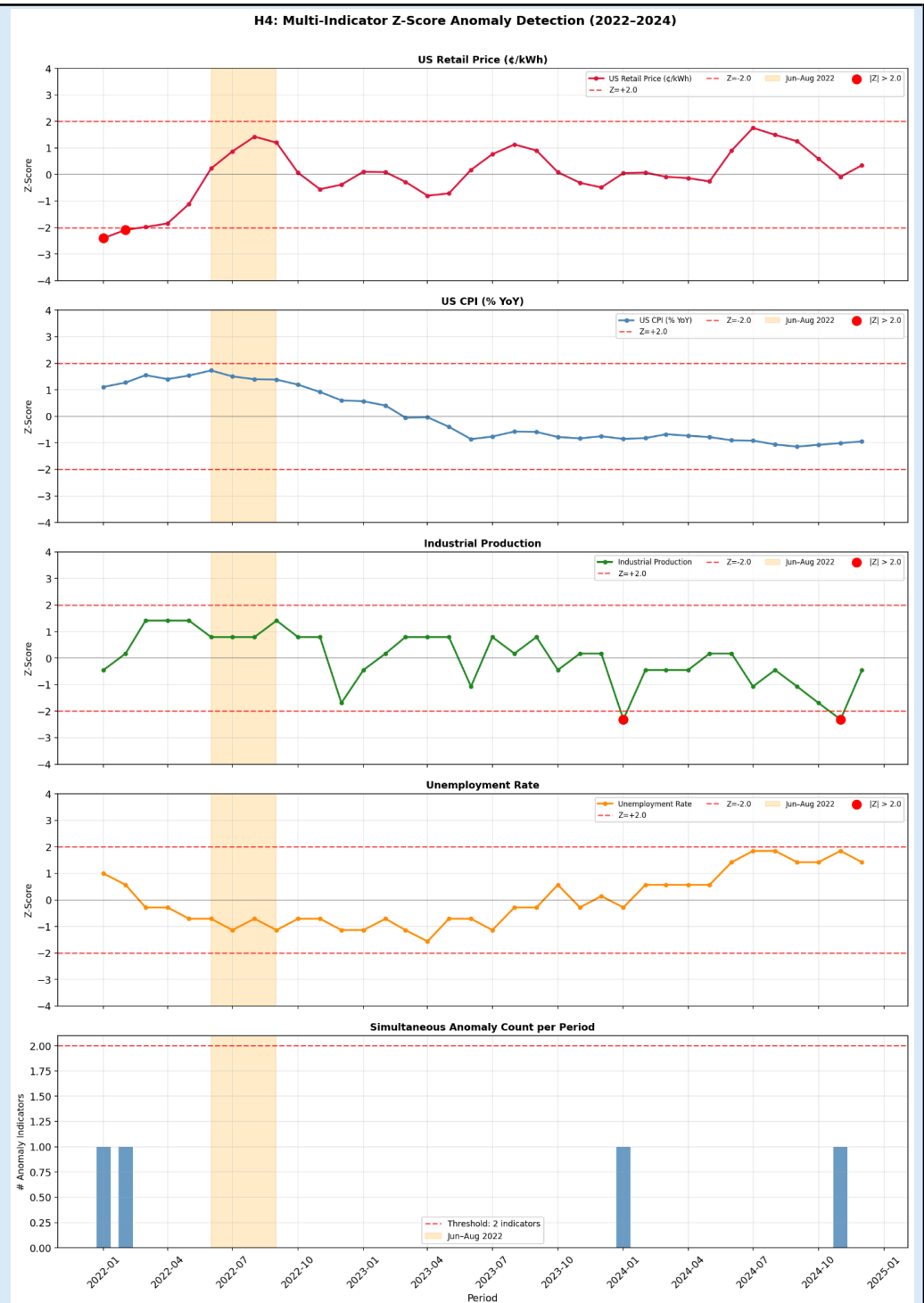


Figure 8. Multivariate anomaly detection and simultaneous indicator breaches based on a $|Z| > 2.0$ threshold framework

Utilizing a multivariate anomaly detection framework within a strict absolute boundary ($|Z| > 2.0$), this line chart tracks the total number of metrics breaching extreme parameters simultaneously within any single given time period. According to the tracking results, although the June-August 2022 window registered immense market volatility, individual indicators (such as CPI and retail Prices) actually breached the strict $|Z| > 2.0$ boundary at slightly shifted, lagging intervals. The total count of concurrent indicator breaches per single month never officially crossed the theoretical anomaly threshold of ≥ 2 variables at once. Instead, isolated extreme anomalies occurred at other independent timestamps, such as a sharp

contraction in the U.S. Industrial Production Index in late 2024 and depressed retail prices in early 2022. This reveals that the transmission of macroeconomic shocks into the energy sector did not manifest as a single synchronized collapse, but rather as a temporally distributed domino effect.

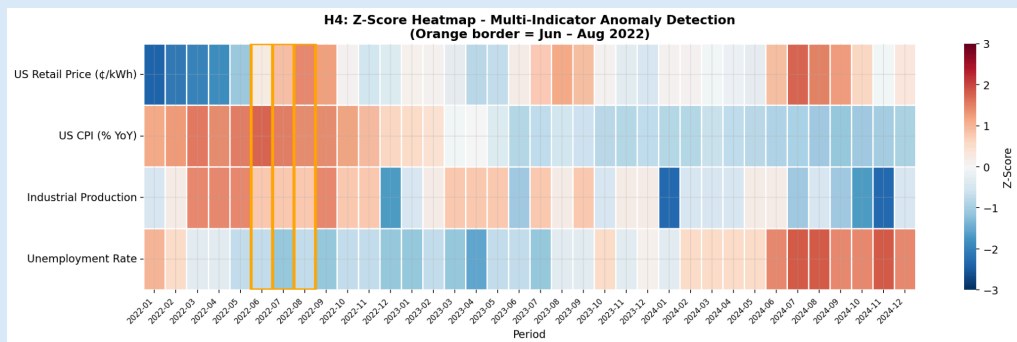


Figure 9. Pearson correlation matrix for all core energy and macroeconomic indicators in the United States

This figure presents a comprehensive Pearson correlation matrix visualizing the pairwise linear relationship among all core energy and macroeconomic indicators within the United States over the 2022-2024 observation period. The matrix employs a color-coded heatmap representation, where red hues indicate positive correlations, blue hues indicate negative correlations, and color intensity reflects correlation strength. Correlation coefficients (r) range from -1.0 (perfect negative linear relationship) to $+1.0$ (perfect positive linear relationship), with values near zero indicating no linear association